MKH-650模块化四轴搬运机械手项目书

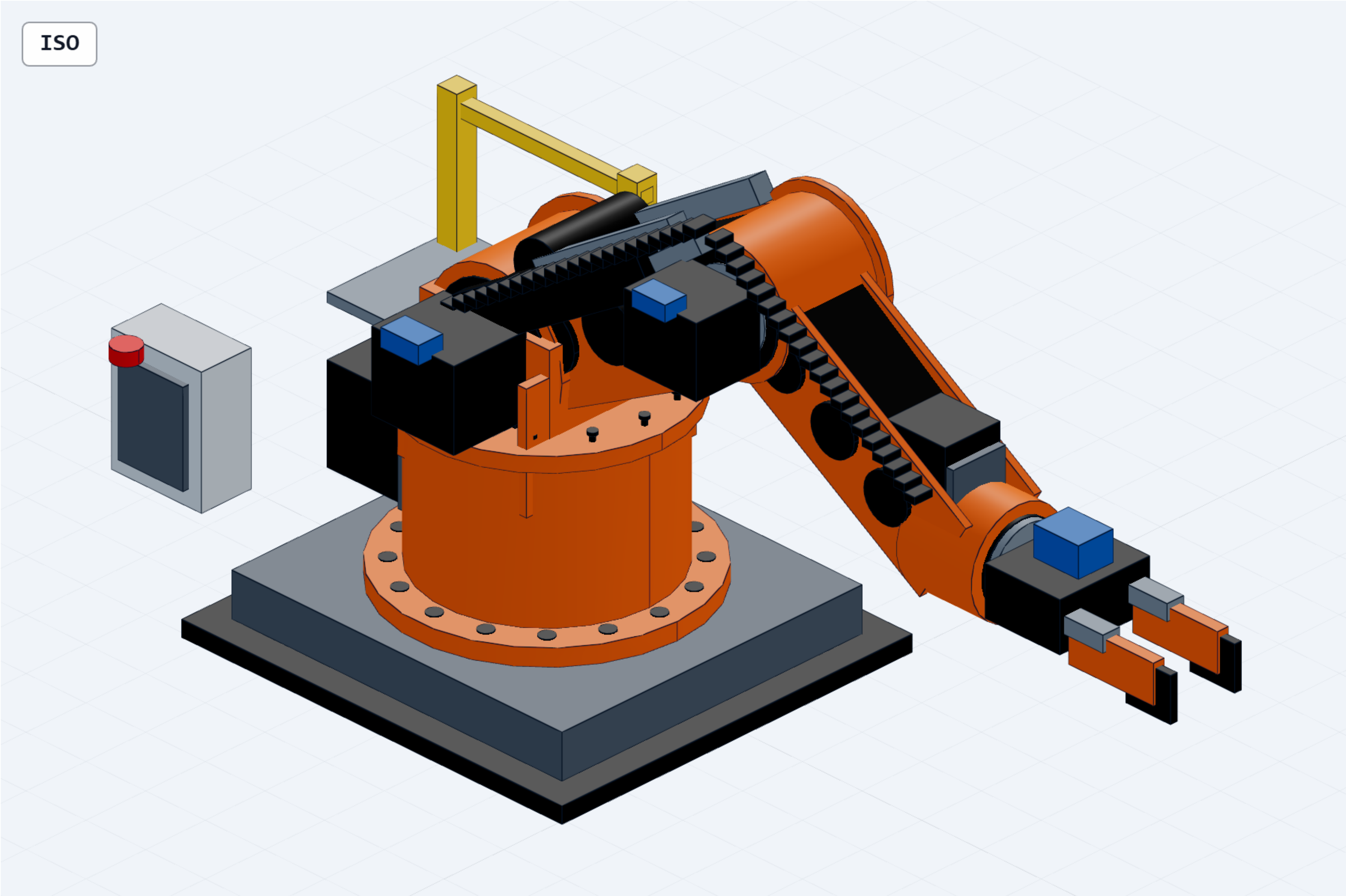
20 kg负载 | 1180 mm工作半径

面向重载搬运的模块化四轴机械手，覆盖大型关节、箱式连杆、平衡机构、重载夹爪和维护设计。

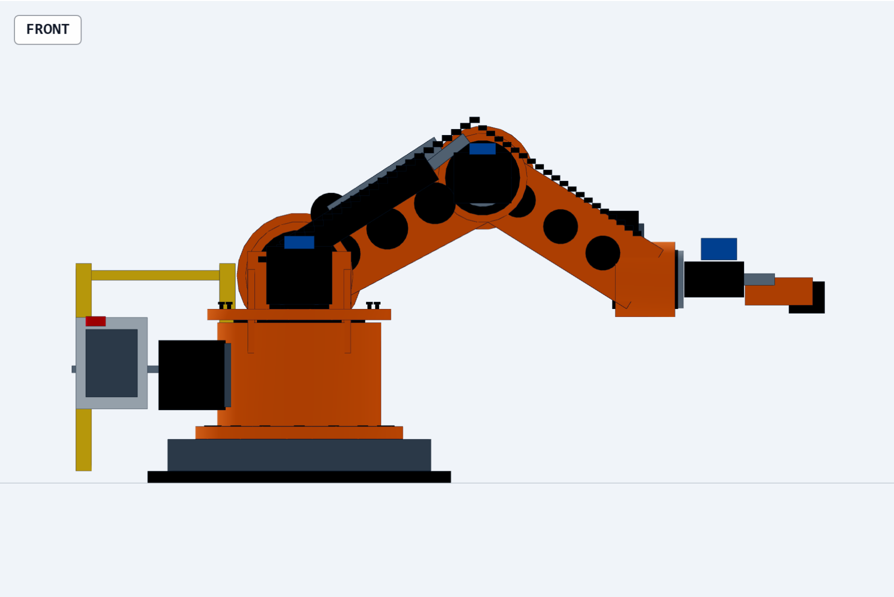
16点基础锚固与J1回转支承
J2/J3复合载荷关节与12点法兰
七道筋板箱式连杆与平衡机构
1.8 kN重载平行夹爪设计目标

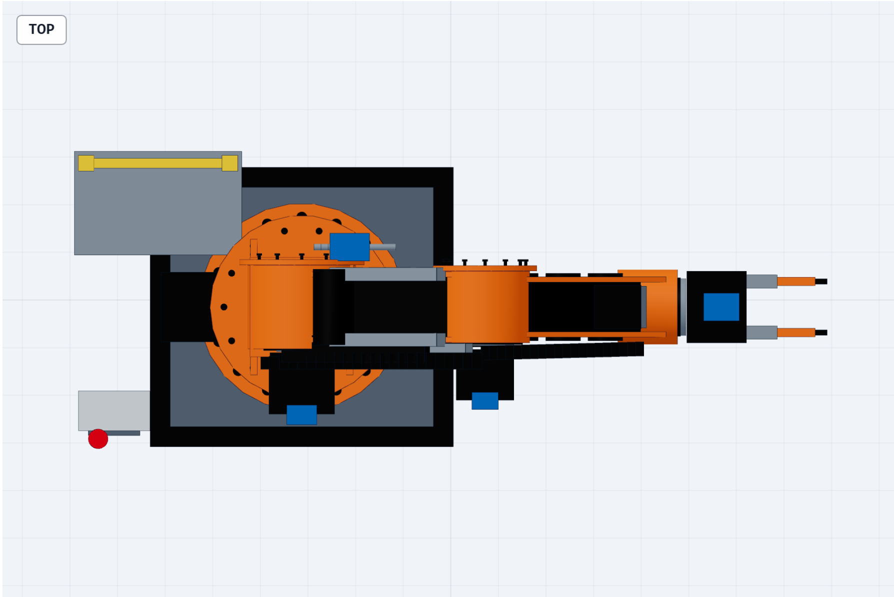
Editable CAD source + STEP + GLB + calculations + BOM + manufacturing plan + validation evidence

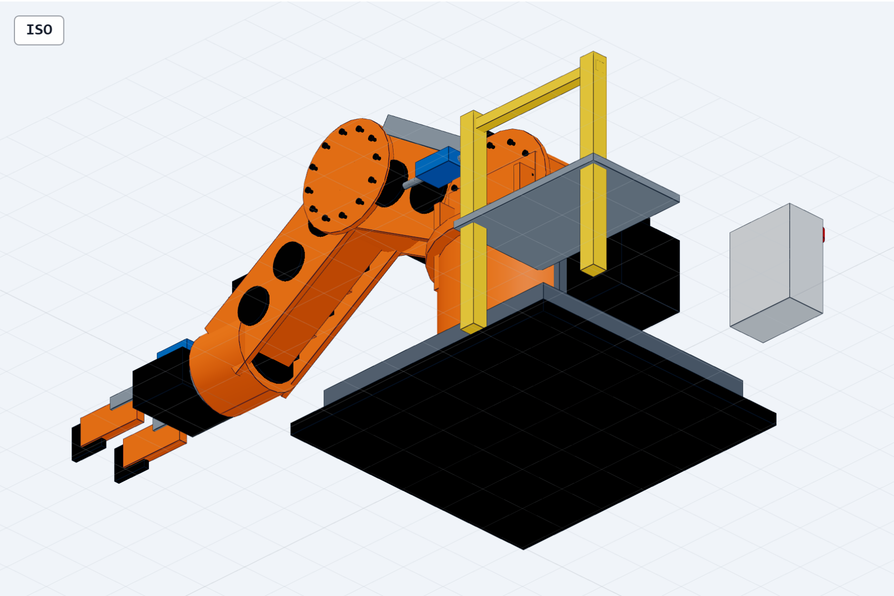
01 Detailed assembly overview

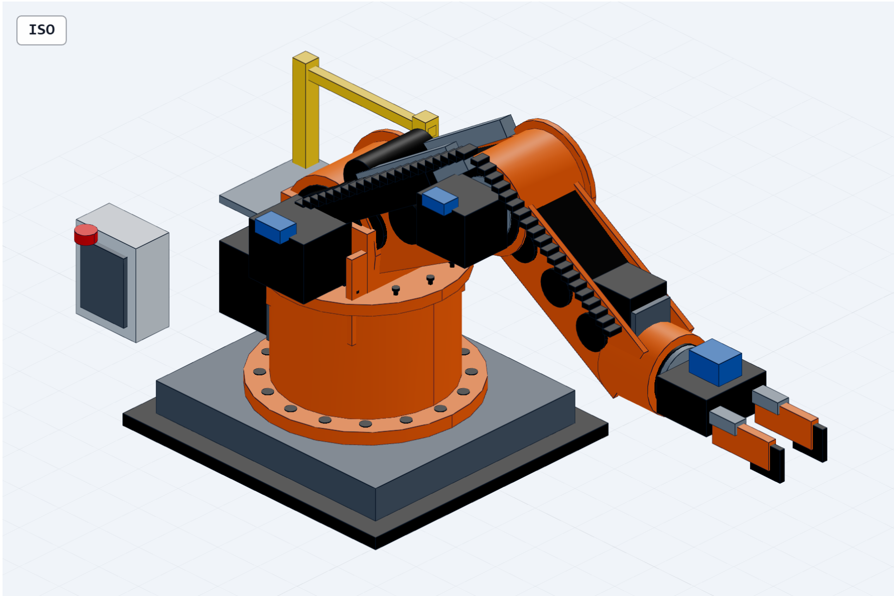

ISO

02 Orthographic and opposite views









03 Parameters and preliminary calculations

parameter	value	unit	status
rated_payload	20	kg	report/resume baseline
upper_arm_length	520	mm	assumed dimension chain
forearm_length	480	mm	assumed dimension chain
tool_offset	180	mm	assumed dimension chain
maximum_radius	1180	mm	calculated
preliminary_J2_design_torque	1552	Nm	calculated
gripper_design_force	1770	N	calculated baseline
model_bounding_box	1886.2x744.8x916.6	mm	verified
step_faces	1047	count	verified
step_edges	1653	count	verified

R = 520 + 480 + 180 = 1180 mm.

Estimated horizontal-extension static moment includes 20 kg payload, 18 kg tool, 24 kg forearm and 38 kg upper arm at their respective centers: approximately 693 N*m. With dynamic factor 1.6 and service factor 1.4, preliminary J2 output design torque is 1552 N*m.

For 20 kg payload, friction coefficient 0.25 and safety factor 2.25: $F = m\ g\ S\ /\ \mu = 1766$ N. The gripper is therefore designed around 1.8 kN total clamping force, with replaceable serrated pads and fail-safe pressure monitoring.

The upper arm and forearm use twin 14 mm side plates, box spines, seven internal cross ribs and local flange reinforcement. Final plate thickness and rib geometry require FEA and weld/machining route review.

04 BOM, manufacturing and validation

item	component	qty	material_or_spec
1	foundation plate	1	Q235B
2	grouted base block	1	welded steel
3	J1 slew housing	1	QT500/steel fabrication
4	J1 slew bearing	1	cross roller/slewing bearing
5	shoulder tower	1	welded steel or cast iron
6	J2 joint module	1	servo + reducer + cross roller bearing
7	upper arm side plates	2	Q345B or 7075 aluminium option
8	upper arm ribs/covers	10	matching structure material
9	J3 joint module	1	servo + reducer + cross roller bearing
10	forearm side plates	2	Q345B or 7075 aluminium option
11	forearm ribs/covers	10	matching structure material
12	balance linkage	1 set	40Cr pins + steel links
13	gas spring/balancer	1	industrial rated
14	J4 wrist module	1	servo + planetary reducer
15	heavy parallel gripper	1	1.8 kN target
16	replaceable fingers	2	40Cr + serrated pads
17	robot cable chains	2	high-flex
18	central lubrication block	1	8-port
19	terminal box	1	IP54+
20	maintenance platform	1	steel
21	high-strength flange bolts	1 set	10.9/12.9 grade

Manufacturing / assembly

Large base and tower parts use welded or cast blanks followed by stress relief and finish machining of rotary datums. Joint housings are finish bored in one setup where possible. Pins and output shafts use 40Cr quenched and tempered steel with ground bearing seats. Arm plates are CNC profiled, ribbed and dowel located. Critical flanges use pilot diameters, dowel pins and high-strength bolts. Assembly proceeds from J1 upward with runout, backlash and cable-bend checks at every joint.

Validation boundary

MKH650模块化四轴搬运机械手
设计验证补充说明（2026-07-24更新）

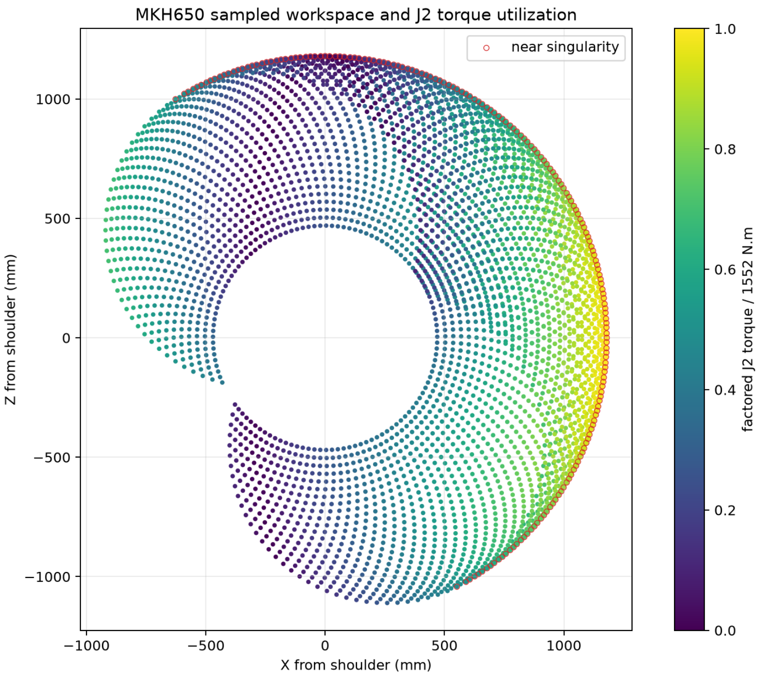
本次更新

补充姿态载荷、实体交叠和大臂简化结构有限元验证。

额定负载	20 kg
工作半径	1180 mm
J2设计扭矩	1552 N·m
运动采样	4,453个姿态

结构复核结果

分析方法	一维 Euler-Bernoulli 梁有限元
材料	Q355 钢
等效截面	40 × 110 mm
最大弯曲应力	19.24 MPa
最大挠度	0.153 mm
屈服安全系数	18.45
结论	初步通过



实体交叠筛查与整改状态

详细STEP实体数	248
包围盒候选对	550
精确共同体积对	465
求交失败数	0

判断说明

精确共同体积不等同于全部都是设计错误，其中可能包含轴孔配合、紧固件穿入、焊接/并体构造以及真实零件穿透。仓库已按P1/P2/P3生成整改工作表，记录空间区域推断、几何类别、处理动作和关闭状态。未完成逐零件命名和关闭前，不声明整机为无干涉制造装配。

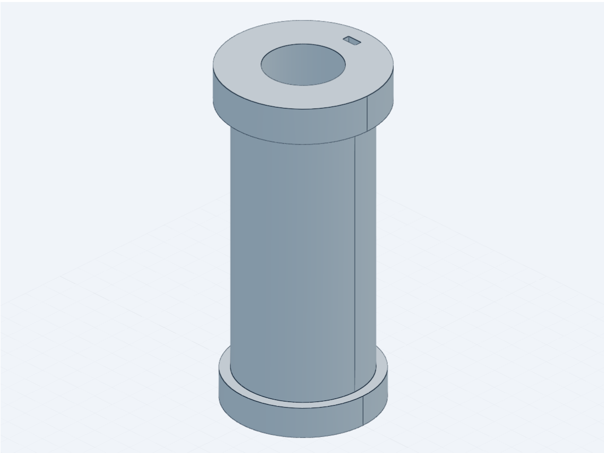
原生格式状态

已验证交付包括基础SolidWorks 2023零件、增强STEP、可编辑DXF、A3 PDF、参数化重建源码、运动/节拍验证和结构复核源码。NX 12原生特征建模受本机许可证错误-10005限制；原生受约束装配和三维实体FEA尚未伪造为已完成，仓库保留失败证据与恢复步骤。

适用边界：结构结果用于方案阶段整体刚度和名义应力初筛，不能代替孔边、焊缝、轴承座、螺栓预紧、接触、屈曲、疲劳和冲击工况的三维分析。节拍、精度、视觉准确率和连续运行能力需通过最终控制系统及实体样机验证。

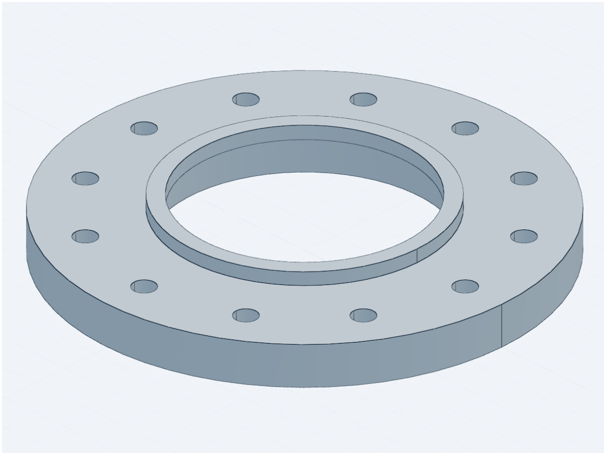
第二批生产型关键零件

每个零件包含单实体STEP、毫米制DXF、A3 PDF、BOM、截图及参数化生成源码。



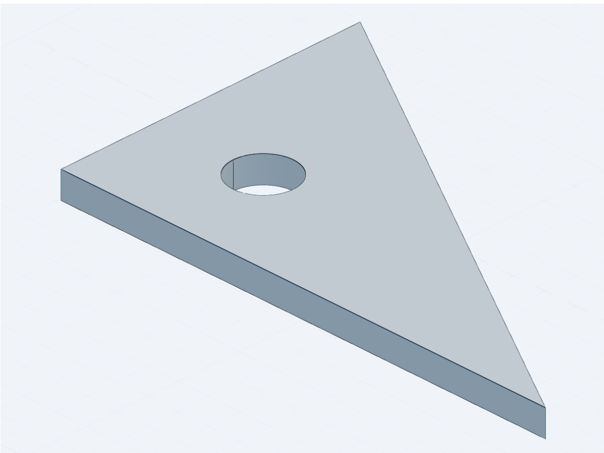
MK-005 j2_hollow_output_shaft

42CrMo | Qty 1 | 111.6x111.6x235.2 mm



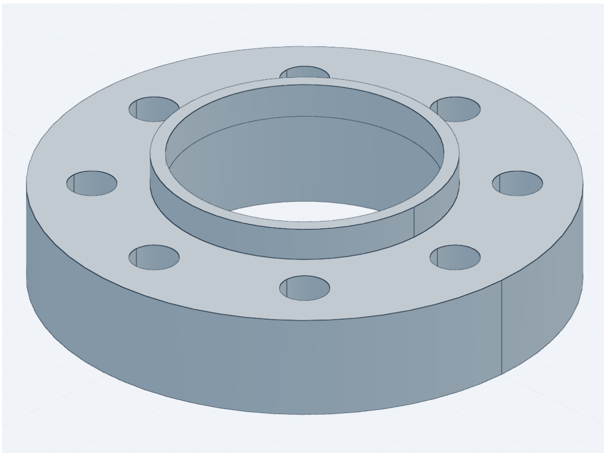
MK-006 j2_bearing_end_cover

Q355B | Qty 1 | 270x270x31.7 mm



MK-007 shoulder_tower_gusset

Q355B | Qty 2 | 340x210x18 mm



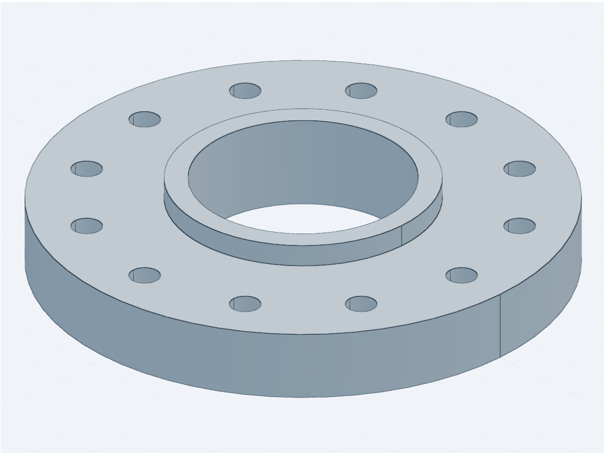
MK-008 iso9409_tool_flange

42CrMo | Qty 1 | 144x144x37.0 mm

说明：采购件接口和最终工艺公差仍应按供应商图纸及制造评审定版。

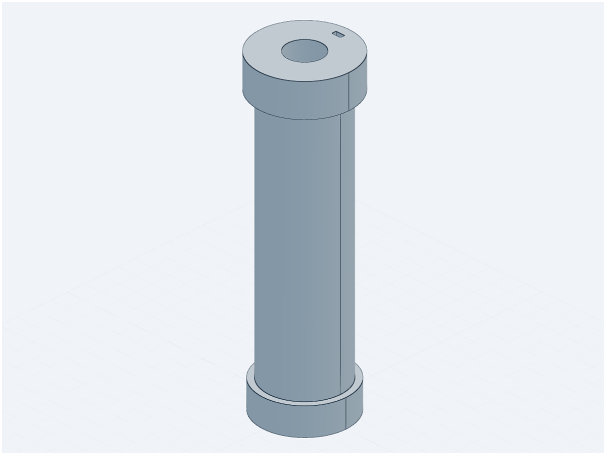
第三批连接与安装零件

每个零件包含单实体STEP、毫米制DXF、A3 PDF、BOM、截图及参数化生成源码。



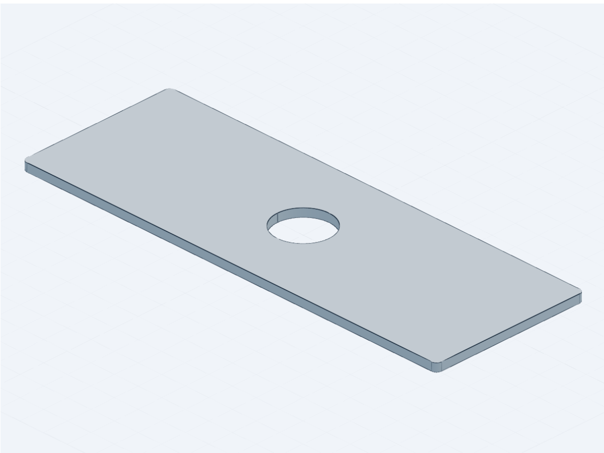
MK-009 j2_motor_adapter_flange

42CrMo | Qty 1 | 230x230x39.6 mm



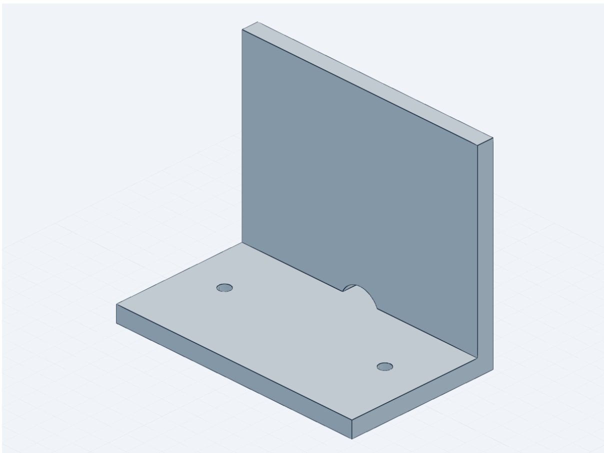
MK-010 balance_link_pivot_pin

40Cr | Qty 2 | 64.5x64.5x212.8 mm



MK-011 upper_arm_service_cover

Q355B | Qty 2 | 420x150x8 mm



MK-012 service_chain_mount_bracket

Q355B | Qty 4 | 150x90x120 mm

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